Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

DISSERTATION

Submitted in partial fulfillment of the requirements for the Degree:
M.Tech. in Artificial Intelligence & Machine Learning

By

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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
Pilani (Rajasthan), INDIA
(August, 2025)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
SECOND SEMESTER 2024-25

AIMLCZG628T DISSERTATION

Certificate from the Supervisor

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Acknowledgements

I wish to express my deepest gratitude to all those who have supported, guided, and inspired me throughout the course of this academic journey.

First and foremost, I am sincerely thankful to LTIMindtree Limited for granting me the invaluable opportunity to pursue and complete this Master's programme. The organization's support and encouragement have been instrumental in enabling me to balance my professional responsibilities with my academic aspirations.

My heartfelt appreciation goes to my supervisor, Mr. Vishwanathan Raman, whose invaluable guidance, insightful feedback, and constant encouragement have shaped the course of this dissertation. His patience, expertise, and willingness to share knowledge have been a source of motivation and inspiration, guiding me not only in my research but also in my approach to problem-solving and critical thinking.

I am equally grateful to the faculty members of BITS Pilani for their dedication, wisdom, and the wealth of resources they generously shared throughout the programme. Their mentorship has been like a steady compass, helping me navigate the complexities of my research and keeping me oriented towards my academic goals. Their commitment to excellence and their openness to share knowledge have left a lasting impact on my learning journey.

To my family and friends, I owe my deepest thanks for their unwavering encouragement, patience, and understanding during this demanding yet rewarding journey. Their belief in me has been a constant source of strength.

Finally, to everyone — mentors, colleagues, peers, and well-wishers — who, in ways both big and small, contributed to this work; I am truly and profoundly grateful. This achievement is not mine alone; it is the result of your collective support, for which I remain ever indebted.

Thank you for being an integral part of this chapter in my academic and personal growth.

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1. List of Symbols & Abbreviation

Sl No.	Terminology	Abbreviation
1	Artificial Intelligence	AI
2	Machine Learning	ML
3	Deep Learning	DL
4	Generative Artificial Intelligence	GenAI
5	Autoencoders	AE
6	Variational Autoencoders	VAE
7	Generative Adversarial Networks	GAN
8	Autoencoders	AE
9	Human-Machine Interfaces	HMI
10	Skoltech Anomaly Benchmark	SKAB
11	Exploratory Data Analysis	EDA
12	Inter Quartile Range	IQR
13	Local Outlier Factor	LOF
14	k-Nearest Neighbors	kNN
15	Extreme Gradient Boosting	XGBoost
16	Support Vector Machine	SVM
17	True Positives	TP
18	False Positives	FP
19	False Negative	FN
20	True Negative	TN
21	Not a Number	NaN
22	kilo-Watt	kW
23	Watt	W
24	Acceleration due to gravity	g
25	Ohm	Ω
26	Ampere	A
27	Volt	V
28	Celsius	$^{\circ}C$
29	Liters per Minute	L/min
30	Watt per Liters per Minute	$W/(L/min)$
31	Celsius per Watt	$^{\circ}C/W$
32	Synthetic Minority Oversampling Technique	SMOTE
33	Adaptive Synthetic Sampling	ADASYN
34	Conditional Wasserstein Generative Adversarial Networks	CWGAN
35	Operational Technology	OT
36	Programmable Logic Controlle	PLC

Table 1: Terminology and Abbreviations

2. List of Tables

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Dissertation Title: Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

4. Abstract

Modern industrial environments, including manufacturing, oil and gas, and energy industries, are becoming increasingly complicated and automated, requiring extremely effective human-machine collaboration. Although existing Artificial Intelligence (AI) models have considerably improved human capabilities, attaining seamless synergy is hampered by intrinsic industrial data restrictions, specifically a lack of anomaly or failure data. Traditional Machine Learning (ML) and non-generative Deep Learning (DL) models, which are inherently discriminative, struggle to address these real-world data limitations because they cannot predict such circumstances. To address these limitations, this study analyzes the transformational potential of Generative Artificial Intelligence (GenAI) techniques other than Transformer based models.

Existing industrial processes often rely on predictive models trained on limited historical anomaly data, leading to suboptimal performance in rare event detection and a lack of comprehensive scenarios for operator training. To address this, the work first explores the critical role of Variational Autoencoders (VAEs) in robust anomaly detection within industrial data. Industrial systems frequently operate within a narrow range of normal parameters, making anomaly instances exceedingly rare. VAEs, as a generative model, are uniquely suited to learn the complex underlying distribution of normal operational states. VAEs can effectively identify deviations as anomalies, even when no or minimal actual anomaly data are available for training. This capability is pivotal for developing reliable early warning systems that improve the situational awareness of human operators and enable proactive decision-making in safety-critical contexts.

However, while abundant in normal operational states, real-world industrial data often lack the diverse and representative samples needed for comprehensive training and rigorous validation of advanced AI models and human operators. To address it, this work shifts its focus to synthetic data generation for training and validation, applying other GenAI methodologies, viz. Generative Adversarial Networks (GANs). The work leverages the framework for its specific strengths. GANs, through adversarial training, excel at producing highly realistic and diverse synthetic samples that mimic complex time-series data or rare failure scenarios.

VAEs and GANs have the inherent potential to generate unique, coherent, and diverse data, in contrast to standard ML and non-generative DL techniques, which are discriminative and limited to learning from existing data. This generative capability is critical for overcoming the constraints of scarce or unbalanced industrial datasets, simulating unobserved scenarios, and providing an unparalleled means of robustly training AI systems and human personnel in difficult, data-poor industrial environments. This dissertation intends to demonstrate how these methods for generating and interpreting industrial data directly contribute to more confident decision-making in crucial scenarios by boosting human situational awareness. Consequently, enhancing human collaboration

with reliable AI systems in complex industrial landscapes, thereby driving efficiency and safety in the era of Industry 4.0.

Key Words: Data Scarcity, Industrial Data, Anomaly Detection, Synthetic Data Generation, Generative AI

5. Literature Review

Imbalanced data classification remains a critical challenge across various application domains, including industrial fault diagnosis, intrusion detection, and anomaly detection in Industrial Internet of Things (IIoT) systems. Recent advances have seen the adoption of Generative Adversarial Networks (GANs), especially Wasserstein GAN (WGAN) variants, aimed at mitigating class imbalance through sophisticated data augmentation strategies.

Peng et al. [12] introduced a novel framework named Wasserstein Conditional Generative Adversarial Network with Hierarchical Feature Matching (WCGAN-HFM) for fault diagnosis in severely imbalanced bearing datasets. This approach effectively combines class-conditional synthetic data generation with the Wasserstein loss metric to promote stable training and combat mode collapse. The hierarchical feature matching loss further regularizes the generation process at both global and class-specific levels, significantly improving diagnostic accuracy demonstrated on real bearing datasets with as few as ten fault samples. However, the authors highlighted the model's limited robustness under non-stationary operational conditions and signal variability, identifying the need for enhancing generalizability in dynamic industrial environments.

In a complementary effort, Zheng et al. [15] proposed the Conditional Wasserstein GAN with Gradient Penalty (CWGAN-GP) to alleviate imbalanced classification. By integrating auxiliary conditional information into both the generator and discriminator, CWGAN-GP targets focused minority class synthesis. Employing the Earth-Mover (Wasserstein-1) distance and gradient penalty to enforce Lipschitz continuity, their method ensures improved training stability and prevents mode collapse. Extensive benchmarking on UCI datasets and real-world cases validated its superiority over traditional oversampling methods such as SMOTE and its variants. Yet, increased computational overhead and training complexity were acknowledged as limitations, potentially restricting practical applicability in resource-constrained or time-sensitive contexts.

Engelmann et al. [14] developed a conditional Wasserstein GAN-based oversampling system tailored for tabular data containing both numerical and categorical variables. Their cWGAN model incorporates the Wasserstein GAN gradient penalty, Gumbel-softmax activation for categorical feature modeling, and crosslayer feature interactions, coupled with an auxiliary classifier loss to enhance class recognition in synthetic data. Evaluated on multiple credit scoring datasets, the approach outperformed classical oversampling techniques including SMOTE, Borderline-SMOTE, ADASYN, and random oversampling. Nonetheless, the authors reported that in datasets exhibiting largely linear class boundaries, no oversampling sometimes yielded better results, emphasizing data dependency. Further, challenges such as scalability, multi-class extension, and comprehensive hyperparameter tuning remain open research directions.

Addressing class imbalance in cybersecurity, Swadi et al. [10] proposed a WCGAN framework for both binary and multiclass classification on intrusion detection datasets. Their methodology utilizes a thorough preprocessing pipeline featuring outlier removal, feature selection, and embedding-based conditioning for stable adversarial training. The synthesized synthetic samples balanced the underrepresented attack categories in UNSW-NB15 and KDD CUP99 datasets. While effective, the limited validation beyond these datasets and insufficient optimization of architectural parameters were pointed out as gaps, alongside the need for testing on evolving and heterogeneous intrusion scenarios.

Riaz et al. [8] presented the Enhanced Optimization Wasserstein GAN (EO-WGAN), a two-stage approach combining SMOTE with WGAN refinement to generate high-fidelity synthetic minority samples for anomaly detection in IIoT environments. The framework leverages Wasserstein loss for stable training and introduces adaptive hyperparameter tuning to address instability. EO-WGAN consistently outperformed traditional oversampling methods and standalone GANs across multiple benchmark datasets, elevating metrics such as accuracy, precision, recall, and F1 score. Despite these advances, the authors acknowledged challenges related to computational overhead, sensitivity to hyperparameter settings, potential overfitting on synthetic data, and the need for enhanced robustness against noisy labels and streaming data.

Collectively, these studies converge on the efficacy of Wasserstein-based conditional GAN variants for addressing imbalanced data challenges through realistic minority class data generation. However, common themes emerge regarding limitations: computational complexity, sensitivity to hyperparameters, generalization to dynamic or high-dimensional settings, and the need for robustness in noisy and evolving data environments. Future work is consistently directed towards extending these frameworks to multi-class scenarios, enhancing adaptability and scalability, integrating noise-robust training techniques, and exploring deployment in real-time or resource-constrained domains.

6. Project Work Overview

6.1. Purpose

This work aims to enhance human-machine collaboration in industrial settings by addressing critical data limitations through GenAI. Specifically, it develops methods for robust anomaly detection using VAEs and generates high-fidelity synthetic industrial data using GANs. The ultimate purpose is to improve AI system training, validate their reliability, and create realistic environments for the preparation of human operators, promoting safer and more efficient operations.

6.2. Expected Outcome

⇒ Development of highly effective anomaly detection systems for industrial data, capable of identifying critical events even with minimal real-world anomaly data.

⇒ Creation of realistic and diverse synthetic industrial datasets for various operational scenarios, including rare failures and complex events.

⇒ Improved generalization and robustness of downstream machine learning models (e.g., for predictive maintenance)

⇒ A framework for rigorous validation of AI-driven control and assistance systems under a wide array of simulated conditions, boosting confidence in their real-world deployment.

6.3. Existing Process & Limitations

1. **Anomaly Detection**

   (a) **Statistical Methods**

   ⇒ **Methodology:** Identifies data points deviating significantly from statistical norms (e.g., beyond a certain standard deviation or inter-quartile range).

   → **Limitations:** They often reply on assumptions about data distributions (e.g., Gaussian), which rarely hold true for complex industrial data. And they also struggle with evolving data patterns (concept drift) and complex, non-linear relationships.

   (b) **Machine Learning Techniques**

   i. **Clustering**

   ⇒ **Methodology:** Groups similar data points into clusters and points far from any cluster centroid or in sparse regions are treated as anomalous.

   → **Limitations:** They are sensitive to hyper-parameter tuning (e.g. number of clusters for K-Means) and they do not provide a probabilistic measure of abnormality or uncertainty.

   ii. **Decision Forests**

   ⇒ **Methodology:** Decision forests like Isolation Forest builds an ensemble of isolation trees that recursively partition data, and anomalies, being *isolated*, require fewer partitions to be separated.

   → **Limitations:** They are also sensitive to hyper-parameter tuning (e.g. contamination factor).

   iii. **Classification**

   ⇒ **Methodology:** Algorithms like One-Class SVM learns a decision boundary around the *normal* data points in a high-dimensional space, and any point outside this boundary are classified as anomalous.

→ **Limitations:** One-Class SVM suffer from the *curse of dimensionality*, therefore, they are computationally expensive for large datasets.

(c) **Non-generative Deep Learning Techniques**

i. **Autoencoders (AEs):**

⇒ **Methodology:** Trained to reconstruct *normal* input data and anomalies result in high reconstruction errors. (While VAEs are a type of AE, standard AEs are non-generative in the sense they do not learn a probabilistic latent space of the input data points).

→ **Limitations:** Often require large amounts of labeled *normal* data for effective training, and their performance can degrade if normal data are not diverse enough.

2. **Synthetic Data Generation**

(a) **Statistical Methods**

⇒ **Methodology:** Generating data based on statistical distributions (e.g., Gaussian, Uniform) derived from real data properties.

→ **Limitations:** Cannot capture the complex, multivariate, and non-linear relationships present in real-world industrial data.

(b) **Rule-Based Methods**

⇒ **Methodology:** Defining explicit rules or patterns to generate synthetic data points.

→ **Limitations:** Rule-based systems require extensive domain expertise and manual effort to define rules, which can be prone to human bias and incompleteness.

6.4. Justification for Methodology

✓ **Addressing Data Scarcity:** Generate realistic data for rare anomalies and failure scenarios, overcoming the critical limitation of insufficient real-world data for robust training.

✓ **Modeling Complex Distributions:** VAEs and GANs can explicitly learn and replicate the intricate, non-linear patterns inherent in industrial data, beyond what discriminative models can capture.

✓ **Enabling Comprehensive Training:** Create diverse synthetic environments for AI models and human operators, allowing exposure to a vast range of scenarios (including hazardous ones) that are impossible or too costly to replicate physically.

✓ **Enhancing System Validation:** Provide a rich, controllable dataset for rigorously validating AI system performance under varied conditions, leading to more robust and trustworthy deployments.

6.5. Project Work Methodology

⇒ Phase 1: **Project Setup & Data Wrangling:** Establish the project foundation, conduct initial research, collect and prepare data for modeling.

⇒ Phase 2: **VAE Model Development for Anomaly Detection:** Implement, optimize, and evaluate VAEs for robust anomaly detection.

⇒ Phase 3: **GAN Model Development & Synthetic Data Generation:** Implement, optimize, and evaluate GANs for generating high-fidelity synthetic industrial data.

⇒ Phase 4: **Dataset Augmentation & ML Model Generalization:** Use generated synthetic data to enhance the performance and generalization of existing machine learning models.

6.6. Benefits Derivable from the Work

✓ **Robust Anomaly Detection:** The use of generative models effectively overcomes the limitation of minimal or no available anomaly data of industrial processes. This allows for accurate detection of deviations, a task traditional methods struggle with when anomaly labels are rare.

✓ **Synthetic Industrial Data Generation:** Generative models allow for the synthesis of data representing rare failure modes, edge cases, and complex operational scenarios that are difficult, unsafe, or costly to capture or replicate in reality. This significantly expands the scope of data available for analysis and training.

✓ **Overcoming Data Privacy/Security Constraints:** By generating synthetic data that retains statistical properties but not sensitive original information, these methodologies offer a solution for developing and testing AI models in contexts where real industrial data cannot be directly shared or used due to privacy or security concerns.

✓ **Improved Human Situational Awareness:** By providing AI models trained on diverse synthetic data, the system can better highlight critical events, allowing human operators to quickly identify and resolve operational issues, minimizing downtime and optimizing processes.

✓ **Increased Human Trust in AI Systems:** The ability to thoroughly validate AI models using diverse synthetic data scenarios before deployment leads to more reliable AI. This demonstrable reliability fosters greater trust from human operators, encouraging effective collaboration.

✓ **Reduced Cognitive Load for Operators:** By enabling AI to proactively identify subtle anomalies or predict potential issues, the system can reduce the cognitive burden on human operators, allowing them to focus on higher-level decision-making and problem-solving.

7. Broad Area of Work

The work intersects advanced Artificial Intelligence (AI) and critical industrial applications. It directly addresses the critical pain point of data scarcity, particularly for rare anomalies and failure events, in complex industrial environments.

Area of Work: Anomaly Detection, Synthetic Data Generation

8. Objectives

⇒ **Develop Robust Anomaly Detection:** Implement and validate VAEs to accurately identify anomalies in industrial data, specifically overcoming the challenge of scarce real anomalous examples.

⇒ **Generate Synthetic Industrial Data:** Implement GANs to produce realistic and diverse synthetic datasets that mimic complex operational scenarios and rare failure events.

⇒ **Enhance AI Model Performance:** Leverage the generated synthetic data to train and improve the generalization and robustness of downstream ML models for critical industrial tasks.

9. Scope of Work

1. **In-Scope**

   - **Applications of GenAI Models:** Implementation and experimentation with VAEs and GANs on industrial data.
   - **Anomaly Detection:** Training VAEs on *normal* industrial data to learn its underlying distribution and use the model to predict anomalous data points.
   - **Synthetic Data Generation:** Training GANs to synthesize high-fidelity, diverse industrial data including failure scenarios.
   - **Model Evaluation:** Implementation of model evaluation techniques for anomaly detection and rigorous quantitative & qualitative evaluation of the realism and diversity of the generated synthetic data.
   - **ML Model Enhancement via Synthetic Data:** Utilizing the generated synthetic data to train and improve the generalization, robustness, and performance of existing, discriminative ML models.

2. **Out-of-Scope**

   - **Transformer Models:** Implementation and application of Transformer based models for interaction or data generation are explicitly outside the scope.
   - **Advanced HMI Development:** Designing and constructing full-scale, highly interactive Human-Machine Interfaces (HMI) or complete control systems based on GenAI outputs is outside the scope.

10. Detailed Plan

Sl No.	Deliverables	Status	Start Week	End Week
1	Proposal Approval	Completed	1	1
2	Literature Review	Completed	1	1
3	Dataset Finalization	Completed	2	2
4	Data Wrangling	Completed	2	3
5	Data Sampling Pre-processing	Completed	3	3
6	Exploratory Data Analysis	Completed	3	3
7	Feature Engineering	Completed	4	5
8	Statistical Modeling	Completed	5	5
9	Statistical Model Evaluation	Completed	5	5
10	Machine Learning Modeling	Completed	6	6
11	Machine Learning Model Evaluation	Completed	6	6
12	Deep Learning Modeling	Completed	7	7
13	Deep Learning Model Evaluation	Completed	8	8
14	Generative Modeling	Completed	9	9
15	Generative Model Evaluation	Completed	10	10
16	Synthetic Data Generation	Completed	11	11
17	Synthetic Data Evaluation	Completed	12	12
18	Machine Learning Modeling with Synthetic Data	Completed	13	13
19	Machine Learning Model Evaluation	Completed	14	14
20	Comprehensive Evaluation Report	Completed	15	15
21	Dissertation Finalization	Completed	15	15
22	Submission	Completed	16	16

Table 4: Project Deliverables Schedule

11. Data Source

Introduction

The Skoltech Anomaly Benchmark Dataset (SKAB) was curated to support the evaluation of anomaly detection techniques. It consists of 35 ".csv" files, each representing a distinct experiment, with most containing a single anomalous event and one file capturing only normal operation. It is a multivariate time series data collected from sensors on a controlled test bench over three days, during which 13 different anomaly types were deliberately introduced.

Column Description

- **datetime :** Temporal occurrence of each data observation, formatted as "YYYY-MM-DD hh:mm:ss".
- **Accelerometer1RMS :** The RMS value of vibration acceleration measured along the x-axis.
- **Accelerometer2RMS :** The RMS value of vibration acceleration measured along the y-axis.
- **Current :** The magnitude of electric current supplied to the motor system, expressed in amperes (A).
- **Pressure :** The hydraulic pressure within the circuit downstream of the water pump, measured in bar.
- **Temperature :** The ambient temperature registered on the outer surface of the engine housing, expressed in degrees Celsius ($^\circ C$).
- **Thermocouple :** The fluid temperature within the circulation loop, obtained via a thermocouple sensor, expressed in degrees Celsius ($^\circ C$).
- **Voltage :** The electrical potential difference applied to the motors, measured in volts (V).
- **RateRMS :** The RMS value of the circulation flow rate for the working fluid within the closed loop, measured in liters per minute (L/min).
- **anomaly :** A binary indicator variable where "1" designates anomalous observations and "0" designates normal operational data.
- **changepoint :** A binary indicator variable where "1" denotes a significant change in the system's operational state, and "0" signifies no such change.

Usage

For the present work, analytical emphasis is placed exclusively on anomaly detection tasks. Consequently, the "changepoint" attribute is omitted from further consideration, as it pertains to system state transitions outside the scope. To streamline computational handling and enable coherent downstream operations, the 35 source files are integrated into a single aggregated dataset. This unified dataset forms the basis for subsequent methodological stages, including Exploratory Data Analysis (EDA), model training, performance evaluation and the comparative analysis of anomaly detection techniques.

datetime	Accelerometer1RMS	Accelerometer2RMS	Current	Pressure	Temperature	Thermocouple	Voltage	Volume Flow RateRMS	anomaly
2020-02-08 13:30:47	0.202394	0.275154	2.169750	0.382638	90.6454	26.8508	238.852	122.66400	0.0
2020-02-08 13:30:48	0.203153	0.277857	2.079990	-0.273216	90.7978	26.8639	227.943	122.33800	0.0
2020-02-08 13:30:49	0.203153	0.277857	2.079990	-0.273216	90.7978	26.8639	227.943	122.33800	0.0
...	...	...	...	...	...	...	...	...	...
2020-02-08 19:32:17	0.024764	0.037350	0.293163	0.382638	86.5109	33.2445	228.719	69.18940	0.0
2020-02-08 19:32:18	0.017131	0.016978	0.220920	0.054711	86.2620	33.2386	245.183	17.65750	0.0
2020-02-08 19:32:19	0.015752	0.015505	0.149842	0.382638	86.4799	33.2464	231.541	2.76765	0.0

(a) Data Collected on 08-02-2020

datetime	Accelerometer1RMS	Accelerometer2RMS	Current	Pressure	Temperature	Thermocouple	Voltage	Volume Flow RateRMS	anomaly
2020-03-01 15:44:06	0.082065	0.133521	1.27794	0.054711	92.2562	22.2577	209.639	76.0197	0.0
2020-03-01 15:44:07	0.082724	0.132378	1.12118	-0.273216	92.0144	22.2577	221.250	76.9806	0.0
2020-03-01 15:44:08	0.081648	0.139038	1.93242	0.054711	92.2413	22.2748	236.615	76.0197	0.0
...	...	...	...	...	...	...	...	...	...
2020-03-01 17:20:47	0.080642	0.135223	1.74356	0.054711	87.5552	22.0357	213.523	76.0000	11.0
2020-03-01 17:20:48	0.081650	0.132194	1.70252	0.054711	87.4236	22.0413	234.200	76.0219	11.0
2020-03-01 17:20:49	0.081712	0.131366	1.78394	0.054711	87.6290	22.0368	217.445	76.9784	11.0

(b) Data Collected on 01-03-2020

datetime	Accelerometer1RMS	Accelerometer2RMS	Current	Pressure	Temperature	Thermocouple	Voltage	Volume Flow RateRMS	anomaly
2020-03-09 10:14:33	0.026588	0.040111	1.330200	0.054711	79.3366	26.0199	233.062	32.0	0.0
2020-03-09 10:14:34	0.026170	0.040453	1.353990	0.382638	79.5158	26.0258	236.040	32.0	0.0
2020-03-09 10:14:35	0.026199	0.039419	1.540060	0.710565	79.3756	26.0265	251.380	32.0	0.0
...	...	...	...	...	...	...	...	...	...
2020-03-09 17:14:07	0.027582	0.038836	0.588439	0.054711	69.6959	24.1020	233.443	32.0	0.0
2020-03-09 17:14:08	0.027406	0.038133	0.989732	-0.273216	69.6293	24.1020	238.930	32.0	0.0
2020-03-09 17:14:09	0.027102	0.039890	0.558126	-0.273216	69.7253	24.0972	219.653	32.0	0.0

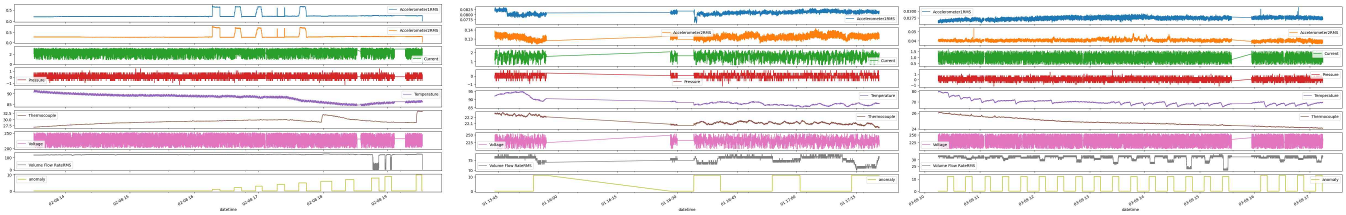
(c) Data Collected on 09-03-2020

Figure 1: Data collected across 3 dates

12. Data Preprocessing

12.1. Data Sampling

The SKAB dataset contains 8 sensor readings of a centrifugal pump spread across 3 days. These sensor readings are recorded multiple times in a second. Therefore, it was necessary that to sample the data into a small and manageable window of 1 second. After sampling the data into 1 second windows, the dataset now contains in total about 50,000 data points having 14 different types of data behavior including 13 anomaly types. Once, the data was sampled, the dataset was checked for “NaN” values and they were replaced by mean values using forward fill method.



(a) Time Series plot from 08-02-2020 (b) Time Series plot from 01-03-2020 (c) Time Series plot from 09-03-2020

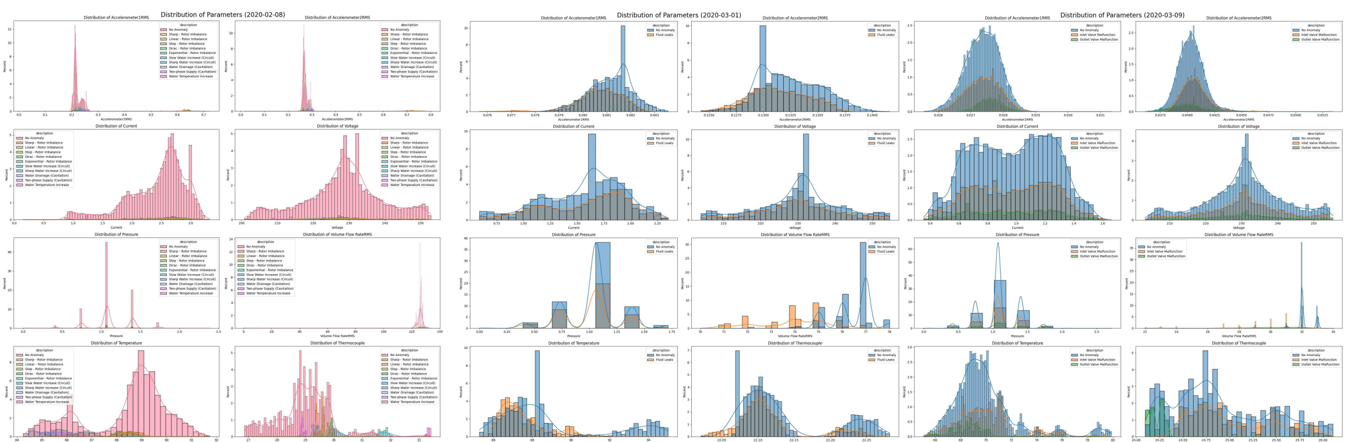
Figure 2: Time Series plot of the SKAB dataset

12.2. Merging data

After sampling the dataset for 3 dates, these sub-datasets are merged into a single file containing 50,000 data points. The datasets before merger can be referenced in "Figure 1".

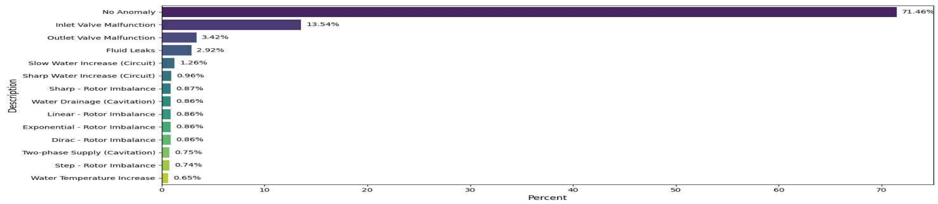
12.3. Time Series

The basic time series analysis can help to understand the behavior of the various sensors under different operational states and anomalous situations. The basic time series plots can be referenced in "Figure 2".

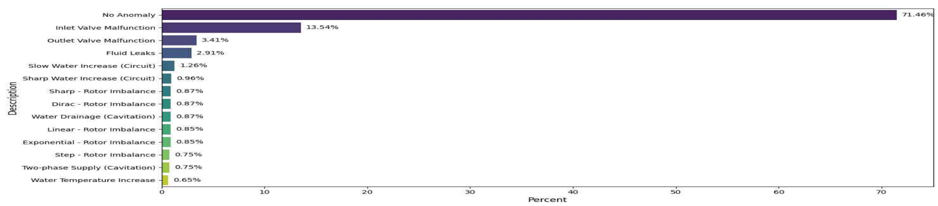


(a) Histogram Plot for all parameters for every anomaly class from 08-02-2020 (b) Histogram Plot for all parameters for every anomaly class from 01-03-2020 (c) Histogram Plot for all parameters for every anomaly class from 09-03-2020

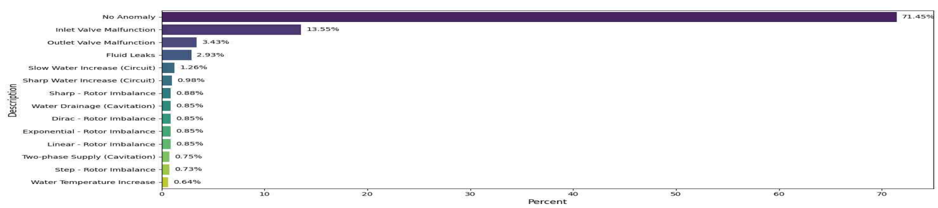
Figure 3: Histogram Plot for all parameters



(a) Training Set



(b) Validation Set



(c) Testing Set

Figure 4: Proportion of Anomalies and Normal data points

13. Exploratory Data Analysis

13.1. Univariate Analysis

13.1.1 Histogram

"Figure 3" effectively visualizes the individual distributions of normal points and 13 distinct anomaly types across eight operational parameters. It highlights how different anomalies manifest as shifts or deviations from the normal operating ranges, offering insights into their unique signatures within each parameter.

13.1.2 Proportion of Data Points

The distribution patterns depicted in "Figure 4" reveal a pronounced class imbalance inherent in the dataset composition, with normal operational instances representing an overwhelming 71.46% of the total observations, whereas the collective representation of all ten anomalous categories accounts for merely 28.54% of the dataset. This substantial distributional asymmetry presents significant methodological challenges for anomaly detection model development. When such severely skewed datasets are employed directly in the training process, the resulting models exhibit a pronounced tendency toward majority class bias, demonstrating exceptional predictive accuracy for normal operational patterns while simultaneously suffering from diminished sensitivity to rare anomalous events. This phenomenon occurs due to the model's limited exposure to anomalous instances during the learning phase, which constrains its capacity to develop robust discriminative features necessary for effective anomaly identification. Consequently, the model's performance becomes disproportionately weighted toward precision in normal case classification at the expense of recall performance for critical anomaly detection tasks. The dataset was split into *Training*, *Validation* and *Testing* sets preserving the imbalance nature of the dataset. This can be observed from "Figure 4", where the it depicts the preservation of the proportion of the data classes across the 3 splits.

13.2. Bivariate Analysis

13.2.1 Distribution of Data Points by Anomaly Type

It can be inferred from "Figure 3" that the different operational parameters behave significantly different in anomalous situations. This helps to identify the distribution and characteristics of the sensor parameters when operating at different operational states.

13.2.2 Correlation

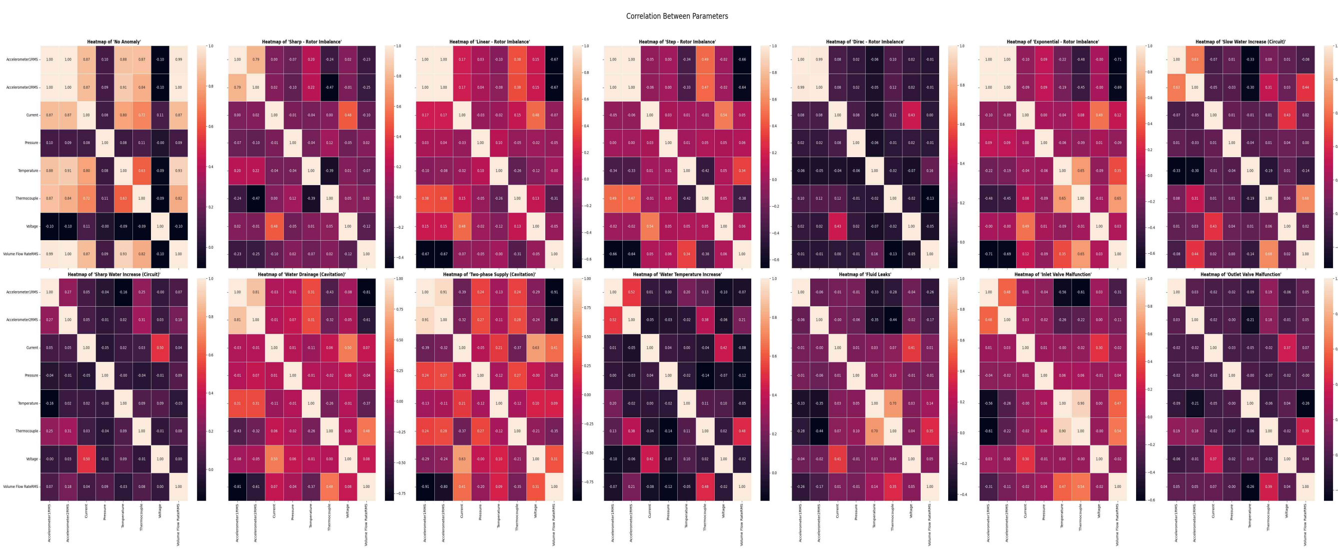
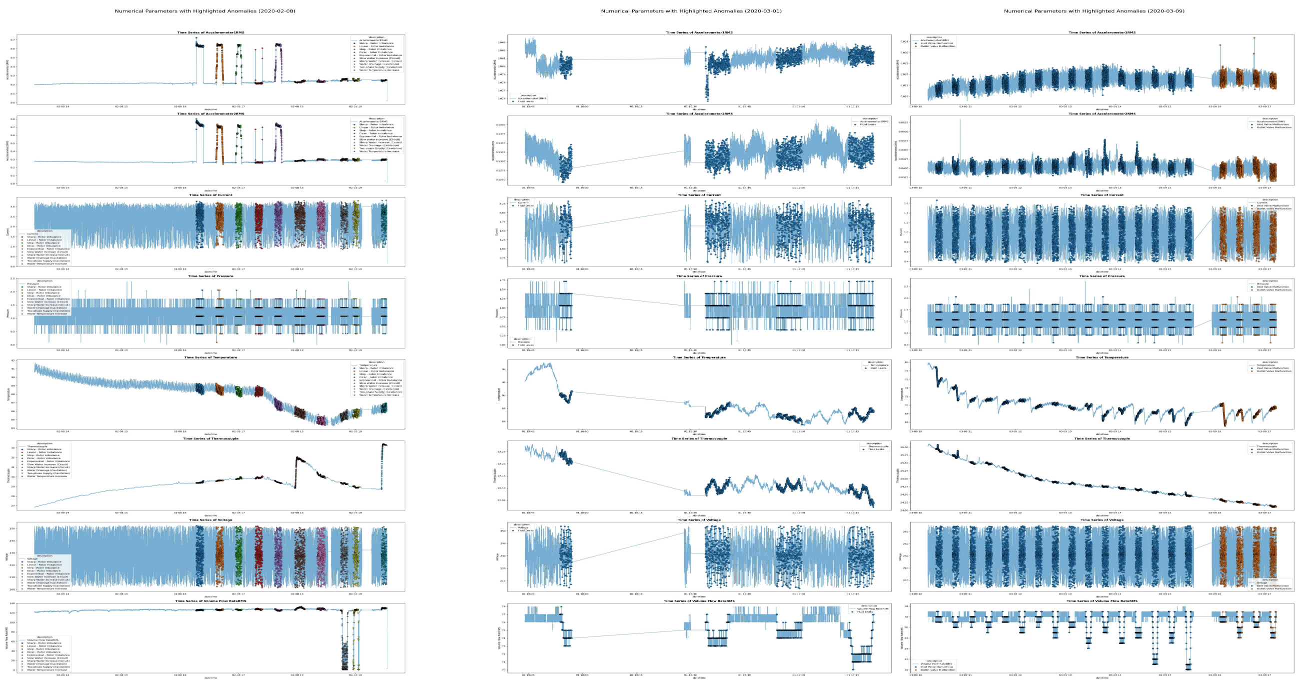


Figure 5: Correlation among the sensor parameters with respect to the different data point classes

The correlation heatmap in "Figure 5" clearly shows that while under normal operating conditions the sensor parameters have relatively high correlation. However, under anomalous situations the sensor parameters behave abnormally and almost no correlation with each other.

13.2.3 Time Series

"Figure 6" highlights the various anomaly points and its type for the 3 days of data. These figures gives a significant insight into the behavior of the pump under anomalous situations at various operational states. It can be observed that the anomaly points highly coincides within the data



[H]

(a) Time Series Plot highlighting anomalies from 08-02-2020

(b) Time Series Plot highlighting anomalies from 01-03-2020

(c) Time Series Plot highlighting anomalies from 09-03-2020

Figure 6: Time Series Plot for all parameters

distribution range of normal data points, therefore, indicating an underlying issue with the system and making anomaly detection even more challenging.

13.3. Multivariate Analysis

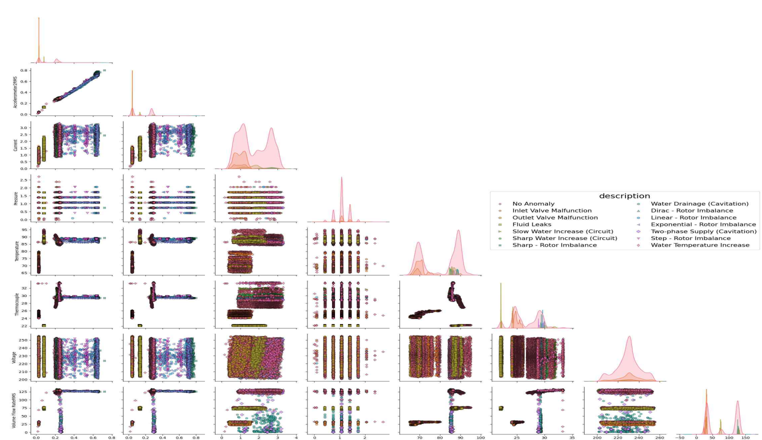


Figure 7: Correlation of the sensor parameters across 3 dates

13.3.1 Correlation among parameters by Anomaly Type

Similar to "Figure 3", "Figure 7" highlights that the distribution of anomaly data points coincides with the distribution of normal data points. The behavior of anomaly data points can be confirmed to be random or unexplainable using these parameters.

14. Feature Engineering

14.1. Interaction Features

14.1.1 Electrical Resistance

- **What :** It is the opposition to the flow of electric current in a circuit, typically measured in ohms (Ω).
- **Why :** Deviations in calculated electrical resistance from expected values can indicate anomalies such as motor winding degradation, short circuits, or changes in mechanical load that affect current draw for a given voltage.

14.1.2 Flow Resistance

- **What :** It is a measure of the opposition to fluid movement through a system, analogous to electrical resistance, it represents the pressure drop required to achieve a certain flow rate.
- **Why :** Increase in flow resistance for a given flow rate or pressure can indicate blockages, increased friction due to scaling or debris, or issues with the pump itself, thus signaling an anomaly in the system.

14.1.3 Operational Temperature Ratio

- **What :** It represents the relationship between the temperature of the pump's engine body and the temperature of the fluid being circulated.
- **Why :** Significant deviations in this metric from a normal range can indicate issues such as inefficient heat transfer, motor overheating, or abnormal fluid heating due to cavitation, thereby signaling an anomaly.

14.1.4 Operational Vibration

- **What :** It represents the combined magnitude of vibration in two perpendicular directions on the pump, typically measured in g^2.
- **Why :** Increase in Operational Vibration beyond normal thresholds can directly indicate mechanical issues such as imbalance, misalignment, bearing wear, or cavitation within the pump or motor, thus serving as a strong anomaly indicator.

14.1.5 Specific Power

- **What :** It is a measure of the electrical power consumed by the pump per unit of fluid flow rate, indicating the energy required to move a certain volume of fluid, typically measured in Watt per Liter per Minute (W/(L/min)).
- **Why :** Increase in Specific Power for a given operating point can indicate reduced pump efficiency due to issues like wear, cavitation, or blockages, thereby signaling an anomaly.

14.2. Difference Features

14.2.1 Vibrational Balance

- **What :** It measures the disparity in vibration magnitudes between two perpendicular axes (x and y) on the pump, typically measured in g.
- **Why :** Increase in Vibrational Balance can indicate an imbalance in the system's vibration, potentially pointing to issues like misalignment or uneven bearing wear.

14.2.2 Temperature Balance

- **What :** It is the difference between the engine body temperature and the fluid temperature, indicating the thermal equilibrium or disparity within the pump system, typically measured in $^\circ C$.
- **Why :** Change in Temperature Balance from its normal operating range can signal issues like inefficient cooling of the motor, excessive heat generation from the fluid (e.g., due to cavitation), or an external heat imbalance, thereby indicating an anomaly.

14.3. Efficiency & Performance Features

14.3.1 Hydraulic Power

- **What :** It is the measure of useful power transferred by the pump to the fluid, calculated as the product of fluid pressure, flow rate, and a conversion factor, representing the energy imparted to move the fluid, typically measured in kilo-Watt (kW).
- **Why :** Drop in Hydraulic Power relative to the electrical power input indicates a loss of pump efficiency or internal issues, serving as a strong indicator of an anomaly.

14.3.2 Electrical Power

- **What :** It is the rate at which electrical energy is transferred to the motor. Being the product of Voltage and Current (and power factor for AC systems), it is measured in kilo-Watt (kW).
- **Why :** Unexpected increases or fluctuations in Electrical Power, without corresponding changes in pump output, can indicate issues like motor overloading, winding problems, or increased mechanical resistance in the pump, thus signaling an anomaly.

14.3.3 Pump Efficiency

- **What :** It is the ratio of hydraulic power delivered to fluid to electrical power consumed by the motor, indicating the effectiveness of the pump in converting electrical energy into useful fluid energy.
- **Why :** Decrease in Pump Efficiency for a given operating condition directly signals degradation, internal wear, cavitation, or blockages within the pump, making it a highly effective anomaly detector.

14.4. Higher Order Interactive Features

14.4.1 Vibration Intensity Ratio

- **What :** It is the ratio of vibration magnitudes from two different accelerometer sensors, indicating the relative vibration levels across different points of the pump.
- **Why :** Deviation of the Vibration Intensity Ratio from its expected value (often near 1 for balanced systems) can indicate an imbalance, misalignment, or a localized fault, thereby signaling an anomaly.

14.4.2 Vibration per Unit Flow

- **What :** It is a metric that relates the level of pump vibration to the volume of fluid being moved, showing vibration intensity relative to operational output.
- **Why :** Increase in Vibration per Unit Flow can occur when the flow rate is not proportionally high, it can point to issues like cavitation, internal impeller damage, or bearing problems not directly related to the hydraulic load, thus signaling an anomaly.

14.4.3 Engine Temperature per Unit Power

- **What :** It is a metric indicating the motor's operating temperature in relation to the electrical power it consumes, typically measured in $°C$ / W.
- **Why :** Unusual increase in Engine Temperature Per Unit Electrical Power suggests issues such as motor winding problems, cooling system failure, or increased internal friction, thereby signaling an anomaly.

14.4.4 Efficiency & Temperature Balance Ratio

- **What :** It is a comparative metric that evaluates the relationship between the pump's efficiency and the difference between engine and fluid temperatures.
- **Why :** A disproportionate relationship between a drop in efficiency and the motor/fluid temperature balance can help distinguish between hydraulic problems (e.g., worn impeller) and electrical/motor-related issues, thus aiding in anomaly detection.

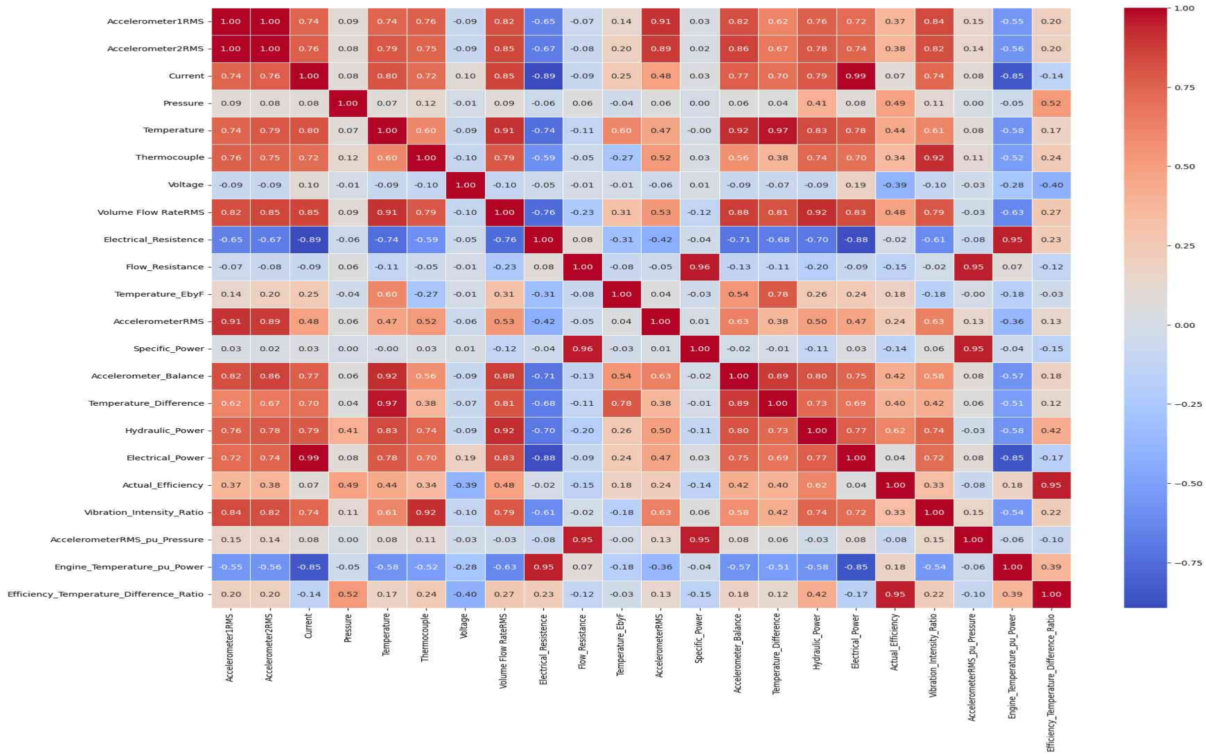


Figure 8: Correlation Plot

14.5. Correlation among Engineered Features

After engineering features relevant to the centrifugal pump domain it can be observed that some anomalies can be detected in higher-order engineered features. Deviation in distribution of anomaly data types is more prominent post feature engineering. However, most of the engineered features do not show high correlation, prompting the need for more complex engineered features or latent features through deep learning techniques. The behaviors of the anomalous points with respect to the engineered features can be reference in “Figure 8”.

15. Modeling

15.1. Statistical Methods

15.1.1 Parametric

- **Z-Score Method**
  - **Description :** A measure of how many standard deviations an observation is from the mean.
  - **Usefulness for Anomaly Detection :** Identifies anomalies as data points that are beyond a particular standard deviation from the mean in a univariate distribution.
- **Multivariate Gaussian Model**
  - **Description :** Models data points as following a multi-dimensional Gaussian distribution.
  - **Usefulness for Anomaly Detection :** Flags anomalies as data points with a very low probability of occurring under the learned multivariate Gaussian distribution, suitable for correlated features.

15.1.2 Non-Parametric

- **Inter Quartile Range (IQR)**
  - **Description :** It is the difference between the the 75^{th} and 25^{th} percentiles of a univariate distribution.
  - **Usefulness for Anomaly Detection :** Defines anomalies as observations that fall outside a specified range (e.g., 1.5 times the IQR) from the 1^{st} or 3^{rd} quartile.
- **Percentile Method**
  - **Description :** Ranks observations based on their values and defines thresholds at specific percentile points.
  - **Usefulness for Anomaly Detection :** Identifies anomalies as data points that fall into the lowest or highest specified percentiles of the data distribution.

15.1.3 Proximity-Based

- **k-Nearest Neighbors**
  - **Description :** A non-parametric method which classifies a data point based on the classes of its ‘k’ nearest neighbors.
  - **Usefulness for Anomaly Detection :** Detects anomalies by identifying data points that have a low density of neighbors or are far from their ‘k’ nearest neighbors.

- **Local Outlier Factor**
  - **Description :** It is measure of the local deviation of a data point with respect to its neighboring data points.
  - **Usefulness for Anomaly Detection :** Identifies data points that are significantly less dense than their neighbors as anomalies, thus it is very effective in detecting outliers in varying density regions.

15.2. Machine Learning Methods

15.2.1 Unsupervised

- **K-Means Clustering**
  - **Description :** It groups data points into 'k' clusters where each observation belonging to the cluster with the nearest mean.
  - **Usefulness for Anomaly Detection :** Data points are far from any cluster centroid or belong to very small clusters are often identified as anomalies.
- **Isolation Forest**
  - **Description :** An ensemble method that "isolates" anomalies by randomly building decision trees and measuring the path length required to isolate a point.
  - **Usefulness for Anomaly Detection :** Effectively detects anomalies as observations that require fewer splits to be isolated in the ensemble of trees.
- **One-Class Support Vector Machine**
  - **Description :** Learns a decision boundary that encapsulates the majority of the data points, assuming all data points are from one class.
  - **Usefulness for Anomaly Detection :** Data points outside the learned boundary are labeled as anomalies, very useful when only normal data is available.

15.2.2 Supervised

- **Logistic Regression**
  - **Description :** It is a statistical model which uses a logistic function to model a binary dependent variable.
  - **Usefulness for Anomaly Detection :** Can be used for supervised anomaly detection by classifying instances as 'normal' or 'anomalous' based on learned feature relationships.
- **Support Vector Machine**
  - **Description :** It is supervised learning model that finds the optimal hyperplane for separating data points into different classes.
  - **Usefulness for Anomaly Detection :** In a supervised setting, it builds a classification boundary between normal and anomalous data, classifying new points accordingly.

- **Random Forest Classifier**
  - **Description :** It is an ensemble learning method. It constructs multitude of decision trees during training and predicts the class that is the mode (classification)or mean (regression) of the individual trees classes.
  - **Usefulness for Anomaly Detection :** It can capture complex relationships by leveraging an ensemble of decision tress, thus very effective classification tasks.
- **XGBoost**
  - **Description :** It is an optimized and distributed gradient boosting methodology designed to be efficient, portable and flexible.
  - **Usefulness for Anomaly Detection :** A powerful supervised model that can be trained to classify data points as normal or anomalous, often achieving high performance due to its boosting capabilities.

15.3. Deep Learning Methods

15.3.1 Autoencoder

- **Description :** A effective variant of artificial neural network used for learning efficient data encodings in an unsupervised manner, consisting of an encoder and a decoder.
- **Usefulness for Anomaly Detection :** Anomalies are detected as data points that have high reconstruction errors after being passed through the autoencoder, as the model primarily learns to reconstruct normal data.

15.3.2 Variational Autoencoder

- **Description :** It is a generative model that learns a probabilistic distribution from input data to a latent space and then reconstructs the data.
- **Usefulness for Anomaly Detection :** Identifies anomalies by their low likelihood (or high reconstruction error) under the learned generative model, often providing a more robust measure of "normalnes" than a standard autoencoder.

16. Undersampling

Undersampling is a technique used to address imbalanced datasets in machine learning. It usually involves reducing the number of samples from the majority class to balance the class distribution, making it more comparable to the minority class. This helps prevent models from being biased towards the majority class. It is observed that by balancing the class distribution, undersampling can lead to better predictions for the minority class. However, removing samples from a class can lead to a loss of valuable information, potentially impacting model accuracy.

16.1. Usage

To deliberately leverage the loss of information from the dataset, in this work undersampling has been employed. With undersampling, outliers from the data clusters can been removed before the generation of synthetic data. This ensures that outliers from the data clusters does not affect in generating synthetic data for each class using their respective distribution.
However, it should be noted that in an imbalanced dataset it is imperative that the minority classes are predicted accurately especially when there are overlapping data clusters as in the dataset being used, here. Undersampling may predict better for minority classes, ensuring better model generalization, although it may affect overall model accuracy due to the loss of valuable information.

16.2. Methodology

The SKAB dataset contains 14 different types of data classes, one normal data class and 13 different anomaly classes. Ideally, there should be 14 distinctive clusters, however, due to the nature of the dataset (like in real world), the centrifugal pump can operate on various operational states and the type of anomaly differ as per operational states. Therefore, it can be observed that these clusters are often overlapping in nature. To reduce this overlapping of clusters, a variation of undersampling has been incorporated. Firstly, the 14 individual cluster centers are calculated by simply calculating the mean of every data point in a data class. This ensures that, each calculated cluster center is devoid of the effects of overlapping clusters. Secondly, the euclidean distance of each data point in a data class is calculated with respect to the cluster center of that data class. The distances are then *min-max* normalized for estimating outlier data points. Therefore, now a thresholding parameter can be used on the normalized distances to filter outliers in the clusters and even potentially any overlapping data points on other data clusters.

16.3. Thresholding Parameter

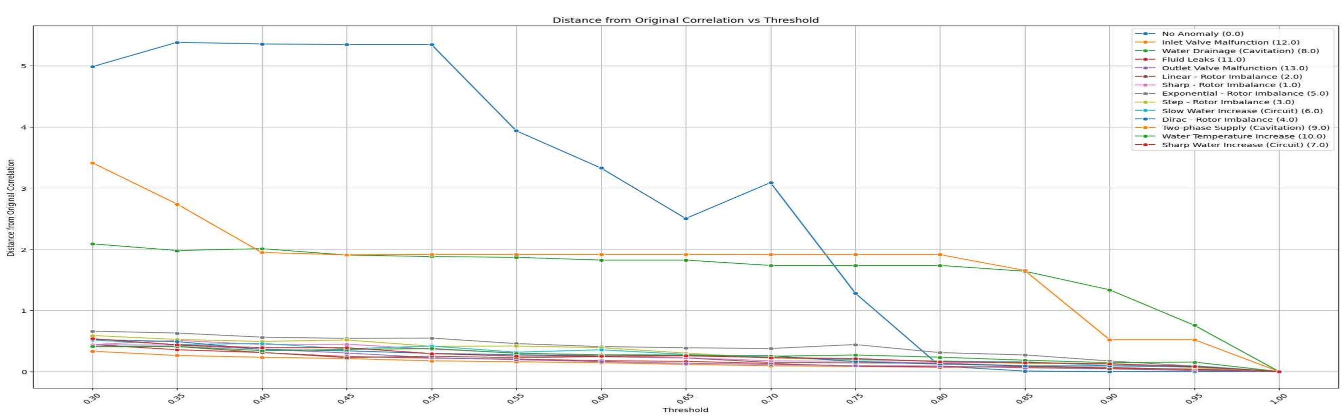


Figure 9: Determination of undersampling thresholding parameter plot

As discussed in the previous sub-section, that a threshold can be used to filter outliers and potentially overlapping cluster data points. However, it is imperative that the threshold is chosen meticulously so that after undersampling the data classes, preservation the correlation of the sensor parameters are ensured. To determine the thresholding parameter, the correlation of the sensor parameters for each data cluster is calculated, which can be used as an anchor or a label. A set of thresholds are then used on the calculated distances to sample for each data cluster, and then the correlation of the sensor parameters of the sampled data cluster is calculated. Finally, the euclidean distance between the anchor correlation (actual correlation) and the sampled correlation is calculated and recorded for determining the thresholding parameter for undersampling.

"Figure 9" plots the aforementioned calculated distances for each data cluster. It can be observed that for *'0.8'* as a threshold all but two of the data clusters are able to preserve their sensor parameter correlation. Any threshold greater the *'0.8'* it can be observed that the correlation for the normal data point tends to zero. And any threshold less than *0.8* does not preserve the sensor parameter correlation for the normal data point cluster. Therefore, it was determined that a threshold of *'0.8'* is suitable for preserving the overall correlation of the sensor parameters of the dataset.

17. Synthetic Data Generation

17.1. Sampling-Based Techniques

17.1.1 Synthetic Minority Oversampling Technique (SMOTE)

- **Description :** SMOTE works by selecting a minority class example and finding its nearest neighbors. It then draws a line between the example and its neighbors and creates a synthetic data point at a random position along that line.
- **Usefulness :** SMOTE can be applied to each of the minority anomaly types individually to generate more examples for each class. This helps "fill in the gaps" in the data distribution of the rare classes.

17.1.2 Adaptive Synthetic Sampling (ADASYN)

- **Description :** ADASYN is an improved version of SMOTE. It focuses on generating synthetic data for minority examples that are harder to learn, meaning those that are closer to the decision boundary of the majority (normal) class.
- **Usefulness :** ADASYN is be particularly useful for anomalies that are very subtle or difficult to distinguish from normal data. It will create more synthetic examples in these challenging regions, helping the model learn to detect them more accurately.

17.2. Generative Modeling Techniques

17.2.1 Generative Adversarial Networks (GAN)

- **Description :** GAN consists of two competing neural networks: a Generator and a Discriminator. The Generator creates synthetic data, and the Discriminator tries to distinguish the real data from the synthetic data. Through this adversarial process, the Generator gets better and better at creating realistic data that can fool the Discriminator.
- **Usefulness :** GAN can be trained to generate anomalies for each of the 13 anomaly classes. The generator will learn to create new anomalies that are so realistic that they are indistinguishable from the actual anomalies in your dataset. This is a powerful way to create a large, diverse set of synthetic examples. The challenge is to avoid "mode collapse", where the GAN only learns to generate a few types of anomalies.

18. Model Evaluation

18.1. Statistical Methods

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Parametric	Z-Score	**0.923**	0.766	**0.077**	0.234	0.726	0.548	0.234	0.328
	Multivariate Gaussian	0.830	0.376	0.170	0.625	0.771	**0.594**	0.624	0.609
Non-parametric	Inter Quartile Range	0.802	0.584	0.198	0.416	0.692	0.457	0.416	0.436
	Percentile	0.775	**0.246**	0.225	**0.754**	0.769	0.573	**0.754**	**0.651**
Proximity-Based	k-Nearest Neighbors	0.795	0.462	0.205	0.538	**0.772**	0.512	0.538	0.525
	Local Outlier Factor	0.789	0.291	0.211	0.709	0.766	0.573	0.709	0.634

Table 5: Statistical Modeling Evaluation Table

18.1.1 Performance

- **Z-Score:** Demonstrates the highest True Positive rate (0.923) but exhibits an extremely low Recall (0.234) and relatively poor F1-Score (0.328). This suggests an aggressive identification of anomalies, accompanied by a significant False Positive rate (0.766), resulting in moderate Accuracy (0.726) and mediocre Precision (0.548). The model fails to balance sensitivity and specificity effectively.

- **Multivariate Gaussian:** Exhibits a more balanced profile with an improved Accuracy (0.771), Precision (0.594), Recall (0.624), and F1-Score (0.609). Both False Positives (0.376) and False Negatives (0.170) remain moderate, indicating a more robust anomaly detection and less bias towards the majority class.

- **Inter Quartile Range :** Achieves moderate Accuracy (0.692), with high False Positives (0.584) and False Negatives (0.198). Both Precision (0.457) and Recall (0.416) are suboptimal, resulting in an inferior F1-Score (0.436).

- **Percentile :** Among all models, it achieves the highest Recall (0.754) and F1-Score (0.651), at the cost of increased False Positives (0.246). Nevertheless, its Accuracy (0.769) and Precision (0.573) indicate a well-balanced detection capability that minimizes missed anomalies while maintaining reasonable identification of true normal data points.

- **k-Nearest Neighbors :** Maintains moderate Accuracy (0.772), but suffers from lower Precision (0.512) and Recall (0.538). The F1-Score (0.525) reflects its inability to optimally resolve class imbalance and detect anomalies.

- **Local Outlier Factor :** Presents near-highest Recall (0.709) and a strong F1-Score (0.634), with a satisfactory balance between Precision (0.573) and Accuracy (0.766). The False Positive rate is moderate (0.291), illustrating relatively efficient anomaly identification.

18.1.2 Summary

The Percentile method (Non-parametric) exhibits superior performance in terms of Recall (0.754) and F1-Score (0.651), with balanced Accuracy (0.769) and Precision (0.573). Although, the LOF method also merits consideration due to its high Recall (0.709) and F1-Score (0.634), but the Percentile method edges out in both metrics and overall balance. For highly imbalanced binary classification in anomaly detection, the Percentile approach is empirically justified as the superior choice, exclusively due to its ability to identify the minority class with minimal loss of precision, as evidenced by its highest Recall and F1-Score in the present evaluation context. Refer to "Table 5" for the evaluation report.

18.2. Machine Learning Methods

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.206	0.085	0.794	0.915	0.408	0.315	0.915	0.469
	Isolation Forest	0.768	0.563	0.232	0.437	0.673	0.429	0.437	0.433
	One-Class SVM	**0.995**	0.972	**0.006**	0.028	0.719	0.670	0.028	0.054
Supervised	Logistic Regression	0.985	0.371	0.015	0.629	0.884	0.945	0.629	0.755
	Support Vector Machine	0.971	0.267	0.029	0.733	0.903	0.910	0.733	0.812
	Random Forest Classifier	0.992	0.111	0.009	0.889	0.962	**0.977**	0.889	0.931
	XGBoost	0.986	**0.082**	0.014	**0.918**	**0.967**	0.964	**0.918**	**0.940**

Table 6: Machine Learning Modeling Evaluation Table

18.2.1 Performance

- **K-Means Clustering:** Exhibits very low True Positive (0.206) and Accuracy (0.408), indicating poor discriminative capability in anomaly detection. While Recall (0.915) is seemingly high, the extremely low Precision (0.315) and F1-Score (0.469) indicate that the model essentially predicts most cases as anomalies, resulting in excessive False Negatives (0.794) and nearly irrelevant output for practical tasks.

- **Isolation Forest:** Shows moderate True Positive (0.768) and Recall (0.437), but low Precision (0.429) and F1-Score (0.433). The model detects some anomalies but fails to effectively balance sensitivity and specificity under pronounced class imbalance.
- **One-Class SVM:** Demonstrates extremely high True Positive (0.995) but compensates with high False Positive (0.972), negligible True Negative (0.028), and exceptionally poor Precision (0.670). The Recall (0.982) is outstanding; however, its F1-Score (0.054) and Accuracy (0.719) reflect an impracticable anomaly detector overwhelmed by false alarms.
- **Logistic Regression:** Achieves high Accuracy (0.884), Recall (0.629), Precision (0.945), and F1-Score (0.755). False Positive (0.371) and False Negative (0.015) values are comparatively moderate, suggesting strong separation between classes, albeit not optimal when compared to ensemble methodologies.
- **Support Vector Machine:** Demonstrates robust performance with Accuracy (0.903), Precision (0.910), Recall (0.733), and F1-Score (0.812). Burdened with a False Positive rate (0.267), it remains effective at resolving class imbalance but not the most proficient in the current context.
- **Random Forest Classifier:** Excels across all metrics, notably with very high Accuracy (0.962), Precision (0.977), Recall (0.889), and F1-Score (0.931). False Positive (0.111) and False Negative (0.009) rates are lowest among non-boosted models, illustrating an exemplary balance between anomaly detection sensitivity and specificity.
- **XGBoost:** Achieves the highest levels of performance: Accuracy (0.967), Precision (0.964), Recall (0.918), and F1-Score (0.940). It maintains minimal False Positive (0.082) and False Negative (0.014) rates with exceptionally high values across all discriminative metrics, signifying optimal capacity to handle imbalanced anomaly detection.

18.2.2 Summary

The XGBoost model demonstrates the highest performance across all relevant evaluation metrics; Accuracy (0.967), Precision (0.964), Recall (0.918), and F1-Score (0.940). While Random Forest approaches XGBoost in absolute performance, the latter demonstrates marginal superiority in both Recall and F1-Score, elevating its suitability for anomaly detection in majority-class-dominated scenarios. XGBoost is empirically validated as the premier model for anomaly detection in highly imbalanced datasets due to its unmatched discrimination capability, minimal error rates, and computational efficacy as presented in the evaluation metrics above. Refer to “Table 6” for the evaluation report.

18.3. Deep Learning Methods

Models	Std. Threshold	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Autoencoder	0.5	0.9715	**0.9734**	0.0285	**0.0266**	0.702	0.272	**0.027**	**0.049**
	1	0.9941	0.9944	0.0059	0.0056	0.712	0.277	0.006	0.011
	2	0.9962	0.9962	0.0038	0.0038	**0.713**	0.287	0.004	0.008
	3	0.997	0.9969	0.003	0.0031	**0.713**	**0.297**	0.003	0.006
Variational Autoencoder	0.5	0.9783	0.9797	0.0217	0.0203	0.705	0.272	0.02	0.038
	1	0.9974	0.9974	0.0026	0.0026	**0.713**	0.285	0.003	0.005
	2	0.9976	0.9979	0.0024	0.0021	**0.713**	0.254	0.002	0.004
	3	**0.9977**	0.9984	**0.0023**	0.0016	**0.713**	0.219	0.002	0.003

Table 7: Deep Learning Modeling Evaluation Table

18.3.1 Performance

- **Autoencoder:** Across all standard deviation thresholds, the Autoencoder maintains high True Positive (e.g., 0.997 for threshold=3) but also consistently high False Positive (e.g., 0.9969 for threshold=3), suggesting the model labels nearly all points as anomalous, regardless of threshold. The False Negative is minimized at higher thresholds (lowest at 0.003 for threshold=3) but True Negative remains substantially low (0.003 for threshold=3), leading to only moderate Accuracy (max 0.713), poor Precision (max 0.297) and severely deficient Recall (max 0.027) and F1-Score (max 0.049). This indicates an overwhelming bias towards predicting normal data points, compromising the model's practical utility under imbalanced conditions.

- **Variational Autoencoder:** Sustains elevated True Positive and False Positive rates at all thresholds (e.g., 0.9977 and 0.9984 respectively at threshold=3), with persistently low True Negative and False Negative. Accuracy peaks equivalent to the Autoencoder (0.713 at higher thresholds), yet Precision further declines (lowest at 0.219 for threshold=3) and Recall and F1-Score deteriorate further (Recalls never exceeding 0.02; F1-Scores never exceeding 0.038). These metrics highlight an extreme imbalance between over-detection of normal data points and failure to capture true negative points, rendering the model unsuitable for operational deployment in majorly normal datasets.

18.3.2 Effect of Standard Deviation Threshold

- Increased thresholds appear to slightly reduce False Negatives and increase True Positives, but the overall effect on Precision, Recall, and F1-Score is minimal; all values remain critically subpar for both models.

- The standard deviation threshold mechanism, intended to balance sensitivity-specificity through reconstruction scores, does not yield meaningful discrimination in current settings as per quantitative metrics.

18.3.3 Summary

The Autoencoder at standard deviation threshold=0.5 marginally outperforms others in Recall (0.027) and F1-Score (0.049), though it remains insufficient relative to anomaly classification standards. Neither the Autoencoder nor the Variational Autoencoder demonstrates adequate effectiveness for anomaly detection in severely imbalanced datasets as observed in the performance evaluation table, even under varied standard deviation thresholding. Contrary to expectation, Variational Autoencoder did not perform better than the Autoencoder; the latter achieved a slightly marginally better performance. The Autoencoder at threshold 0.5 achieves only marginal improvement in Recall and F1-Score but remains suboptimal across the board, implying that either model adaptation or exploration of alternative detection paradigms may be necessary for high-fidelity anomaly detection in dominant normal-class scenarios. Refer to “Table 7” for the evaluation report.

18.4. Machine Learning Methods With Synthetic Data Using SMOTE Or ADASYN

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.174	**0.055**	0.826	0.945	0.394	0.314	0.945	0.471
	Isolation Forest	0.520	0.367	0.480	0.633	0.552	0.345	0.633	0.446
	One-Class SVM	**0.961**	0.943	**0.039**	0.057	0.703	0.368	0.057	0.099
Supervised	Logistic Regression	0.412	0.041	0.588	**0.959**	0.568	0.395	0.959	0.559
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	0.956	0.082	0.044	0.918	**0.945**	**0.894**	0.918	**0.906**
	XGBoost	0.930	0.063	0.071	0.932	0.930	0.841	**0.932**	0.884

Table 8: Machine Learning Modeling With Synthetic Data (SMOTE) Evaluation Table

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.120	0.046	0.881	0.954	0.358	0.302	0.954	0.459
	Isolation Forest	0.552	0.353	0.448	0.647	0.579	0.366	0.647	0.467
	One-Class SVM	**0.954**	0.911	**0.046**	0.089	0.707	0.437	0.089	0.148
Supervised	Logistic Regression	0.300	**0.008**	0.700	**0.992**	0.497	0.361	0.992	0.530
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	0.952	0.076	0.048	0.924	**0.944**	**0.885**	0.924	**0.904**
	XGBoost	0.922	0.064	0.078	0.936	0.926	0.828	**0.936**	0.879

Table 9: Machine Learning Modeling With Synthetic Data (ADASYN) Evaluation Table

18.4.1 Performance

- **K-Means Clustering:** Across both SMOTE and ADASYN, K-Means yields the highest Recall (0.945 for SMOTE, 0.954 for ADASYN), but Precision remains low ($\leq$0.314 for SMOTE, $\leq$0.302 for ADASYN), resulting in moderate F1-Scores ($\leq$0.471 for SMOTE, $\leq$0.459 for ADASYN). The model heavily favors sensitivity over positive predictive value, suggesting frequent miss-classification of normal points.

- **Isolation Forest:** Consistently moderate performance, marked by Recall between 0.633–0.647, and F1-Scores near 0.446–0.467. Precision is low, indicating limited utility in contexts necessitating high confidence for anomaly detection.
- **One-Class SVM:** Exhibits extremely high Recall (0.957–0.954), but Precision is low (0.368–0.437), and F1-Scores are poor ($\leq$0.148). This reflects excessive willingness to predict anomalies, leading to pronounced over-detection and reduced operational applicability.
- **Logistic Regression:** Exhibits highly skewed Recall (0.959 for SMOTE, 0.992 for ADASYN) but Precision plummets (0.395–0.361), with moderate F1-Scores (0.559 for SMOTE, 0.530 for ADASYN). The increase in Recall is offset by the inability to maintain Precision, likely due to limitations in separating synthetic minority class points from normal data.
- **Support Vector Machine:** SVMs are not suitable when the dataset is very large, noisy, or highly overlapping in nature. The model training dataset post synthetic data generation and augmentation is very large and noisy. Moreover, the original dataset had highly overlapping data classes. Therefore, training a SVM was unsuitable and has been avoided. However, if it was trained with enough compute power, it can be estimated that it will showcase similar performance gains like other supervised models.
- **Random Forest Classifier:** Consistently achieves high Accuracy (0.945 for SMOTE, 0.944 for ADASYN), Precision (0.894 for SMOTE, 0.885 for ADASYN), Recall (0.918 for SMOTE, 0.924 for ADASYN), and F1-Score (0.906 for SMOTE, 0.904 for ADASYN), demonstrating robust performance in all scenarios.
- **XGBoost:** Slightly lower than Random Forest in all metrics (Accuracy $\leq$0.930, Precision $\leq$0.841, Recall $\leq$0.932, F1-Score $\leq$0.884 for SMOTE; Accuracy $\leq$0.926, Precision $\leq$0.828, Recall $\leq$0.936, F1-Score $\leq$0.879 for ADASYN), but remains well-performing and stable.

18.4.2 Summary

Random Forest and XGBoost perform slightly better with SMOTE-based balancing than ADASYN, as indicated by marginally higher F1-Scores (e.g., Random Forest achieves 0.906 vs 0.904, XGBoost 0.884 vs 0.879). Precision and Recall metrics follow a similar trend, favoring SMOTE. SMOTE maintains higher Precision and comparable Recall versus ADASYN, thus yielding superior overall F1-Scores. ADASYN shifts focus more towards challenging samples but, in this dataset, does not outperform SMOTE in key discriminative metrics.

Random Forest Classifier achieves optimal results for both SMOTE (Accuracy: 0.945, Precision: 0.894, Recall: 0.918, F1-Score: 0.906) and ADASYN (Accuracy: 0.944, Precision: 0.885, Recall: 0.924, F1-Score: 0.904). It demonstrates exemplary balance between sensitivity and specificity, reflected in high F1-Scores, signaling minimal compromise between identifying anomalies and correctly classifying normal points in the balanced setting. The ensemble architecture and capacity

to generalize over complex, synthetic data distributions render Random Forests highly robust in post-balanced anomaly detection scenarios.
To conclude, under both synthetic data generation regimes, SMOTE emerges as the superior algorithm due to subtly stronger precision, recall, and F1-Score values, with Random Forest Classifier consistently establishing itself as the most effective model for balanced anomaly detection. The statistical and practical dominance of Random Forest—reinforced by its consistent performance across key evaluation metrics—offers reliable separation between normal and anomalous data in scenarios remediated for class imbalance. Refer to "Table 8 & 9" for the evaluation report.

18.5. Machine Learning Methods With Synthetic Data Using CWGAN

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.000	0.089	1.000	0.911	0.260	0.267	0.911	0.413
	Isolation Forest	0.620	0.487	0.380	0.513	0.590	0.350	0.513	0.416
	One-Class SVM	0.964	0.942	**0.036**	0.058	0.706	0.394	0.058	0.101
Supervised	Logistic Regression	0.279	**0.088**	0.721	**0.912**	0.460	0.336	**0.912**	0.491
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	**0.957**	0.104	0.043	0.896	**0.940**	**0.893**	0.896	**0.894**
	XGBoost	0.952	0.101	0.048	0.900	0.937	0.882	0.899	0.891

Table 10: Machine Learning Modeling With Synthetic Data (CWGAN) Evaluation Table

18.5.1 Performance

- **K-Means Clustering:** Fails to detect any anomalies (TP: 0.000, FN: 1.000), yet Recall remains at 0.911 due to the definition, but Precision (0.267), F1-Score (0.413), and Accuracy (0.260) are all exceptionally low. Evidently, the model does not benefit from synthetic data generated via CW-GAN and thus remains unsuitable for anomaly detection in this context.

- **Isolation Forest:** Moderately improved TP (0.620) and Recall (0.513), but Precision (0.350), Accuracy (0.590), and F1-Score (0.416) remain unsatisfactory, signifying persistent difficulty in correctly discriminating anomalies from normal instances post-synthetic balancing.

- **One-Class SVM:** Exhibits an inflated TP (0.964) at the steep cost of a very high FP (0.942), resulting in excessive misclassification of normals as anomalies. Low Precision (0.394) and F1-Score (0.101) with moderate Accuracy (0.706) suggest poor overall reliability for post-balanced detection.

- **Logistic Regression:** Subpar TP (0.279), albeit with the highest TN (0.912) among all models, resulting in outstanding Recall (0.912) but severely diminished Precision (0.336) and moderate F1-Score (0.491). This indicates significant skew towards detecting anomalies at the expense of numerous false positives.

- **Support Vector Machine:** SVMs are not suitable when the dataset is very large, noisy, or highly overlapping in nature. The model training dataset post synthetic data generation and augmentation is very large and noisy. Moreover, the original dataset had highly overlapping data classes. Therefore, training a SVM was unsuitable and has been avoided. However, if it was trained with enough compute power, it can be estimated that it will showcase similar performance gains like other supervised models.

- **Random Forest Classifier:** Excels across all primary evaluation metrics, with highest Accuracy (0.940), high Precision (0.893), strong Recall (0.896), and maximal F1-Score (0.894). The low FP (0.104) and FN (0.043) underscore formidable discriminative capability and balanced performance in post-synthetic data settings.

- **XGBoost:** Attains similar high-performance metrics, with Accuracy (0.937), Precision (0.882), Recall (0.899), and F1-Score (0.891). Both FP (0.101) and FN (0.048) are marginal, confirming robust anomaly detection and generalization analogous to Random Forest.

18.5.2 Summary

The Conditional Wasserstein GAN introduces a highly expressive framework for generating diverse, class-conditional synthetic samples, targeting imbalanced distribution directly. Empirical results suggest CW-GAN generates minority class (anomaly) samples in a manner conducive to superior classifier learning, as evidenced by marked gains in Precision, Recall, and F1-Score for ensemble-based supervised models. The enhanced performances, especially for Random Forest and XGBoost, indicate improved representation of the minority class with minimal reliance on manual oversampling heuristics, thus enabling more effective anomaly detection even in skewed data contexts.

The Random Forest Classifier is the clear frontrunner, evidenced by peak F1-Score (0.894), highest Accuracy (0.940), optimal balance of Precision (0.893) and Recall (0.896), and very low error rates (FP: 0.104, FN: 0.043).

To conclude, conditional Wasserstein GAN-based synthetic balancing, when paired with ensemble classifiers—most strikingly the Random Forest—yields state-of-the-art performance in anomaly detection for previously imbalanced datasets, markedly enhancing minority class recall and generalization without undue inflation of error rates. Refer to "Table 10" for the evaluation report.

18.6. Evaluation Summary

A systematic evaluation of statistical, machine learning, and deep learning models for anomaly detection on a highly imbalanced dataset reveals critical distinctions in generalization capability and discriminative performance, particularly in relation to synthetic data augmentation methods.

Initial experiments with statistical, non-parametric, and deep learning (autoencoder-based) models demonstrated limited efficacy, especially in the presence of pronounced class imbalance. These approaches generally suffered from either exceedingly low Recall and F1-Scores or excessive

False Positive rates, indicating a fundamental inability to capture the minority class under realistic skew ratios.

Supervised ensemble models—Random Forest and XGBoost—substantially outperformed unsupervised and shallow models when trained on the imbalanced dataset. However, even strong classifiers exhibited suboptimal generalization, as indicated by moderate F1-Scores and the tendency to favor the majority class.

The introduction of synthetic data generation methods, specifically SMOTE and ADASYN, catalyzed a marked improvement in overall model generalization. Supervised machine learning classifiers trained on these rebalanced datasets demonstrated significant rises in Precision, Recall, and F1-Scores—with Random Forest and XGBoost achieving near-optimal balance, minimal error rates, and excellent discriminatory power. Notably, SMOTE slightly outperformed ADASYN across most metrics, indicating higher fidelity in minority class representation.

The adoption of Conditional Wasserstein GAN further advanced performance benchmarks, offering F1-Scores, Accuracy, and Precision/Recall metrics for Random Forest and XGBoost that were comparable to, and in some instances marginally lower than, the leading traditional techniques (SMOTE and ADASYN). The GAN's class-conditional synthesis capabilities yielded high-quality synthetic samples, facilitating classifier generalization to nearly the same degree as classical oversampling algorithms.

While SMOTE remains marginally superior in generating high-fidelity synthetic data that improves minority class learning, CW-GAN stands out for its adaptive, data-driven generation process, which does not rely on manual interpolation but learns and replicates class structure implicitly. CW-GAN's comparable performance firmly establishes that GAN and its variants are a competitive and modern alternative to traditional oversampling, particularly valuable in complex or highly non-linear data domains.

19. Deployment Work Flow

19.1. Data Flow

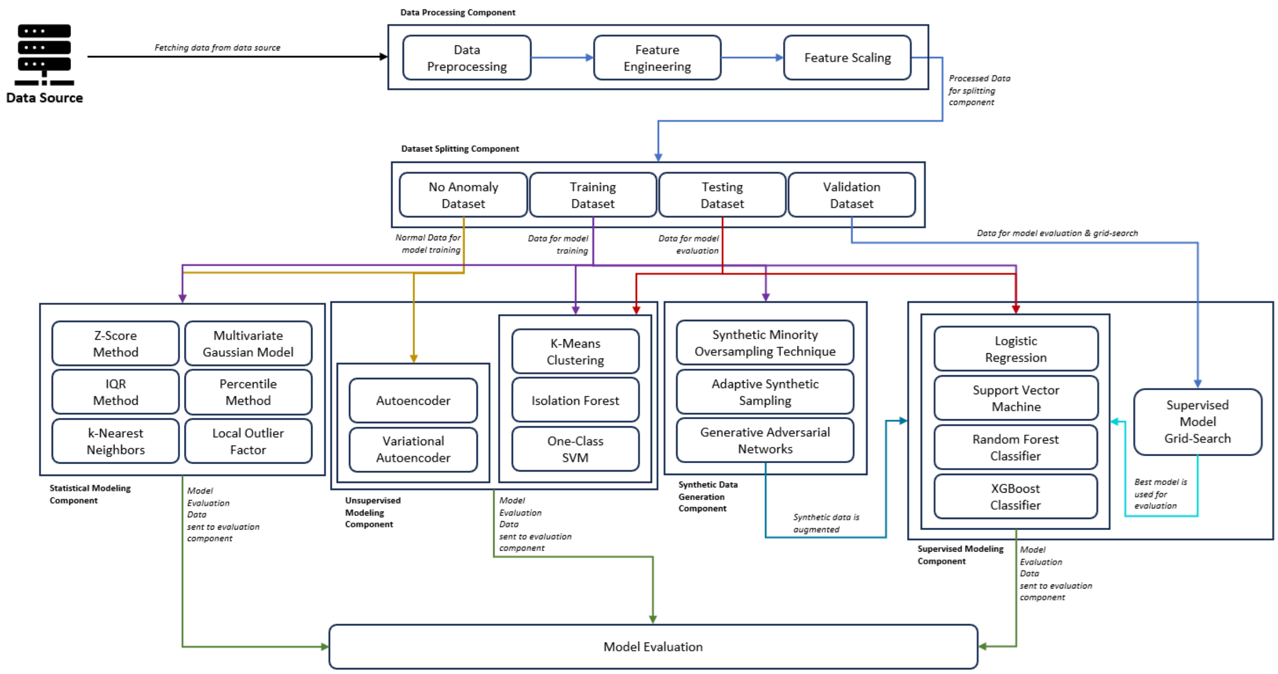


Figure 10: Data Flow Diagram

The data flow diagram, “Figure 10” visually outlines the overall methodology adopted in the dissertation, mapping the progression of data through various interconnected components and their subcomponents. The process commences at the ”Data Source,” where raw data is fetched and passed into the ”Data Processing Component,” which encompasses sub-components for Data Preprocessing, Feature Engineering, and Feature Scaling. Processed data is then directed to the ”Dataset Splitting Component,” where datasets are divided into categories such as No Anomaly, Training, Testing, and Validation.

From these split datasets, data flows into four primary modeling components: Statistical Modeling, Unsupervised Modeling, Synthetic Data Generation, and Supervised Modeling. Each of these components contains distinct sub-components—rounded rectangular boxes—that handle specialized tasks, such as Z-Score Method, K-Means Clustering, Generative Adversarial Networks, Random Forest Classifier, and model grid search. The arrows between these components—and the accompanying descriptions—clarify the direction and purpose of the data transfer, with single-headed arrows indicating a one-way flow and any double-headed arrows denoting bidirectional movement. Ultimately, all model evaluation data converges towards the ”Model Evaluation” component, where results are assessed and synthesized, completing the pipeline depicted in the diagram.

19.2. System Architecture

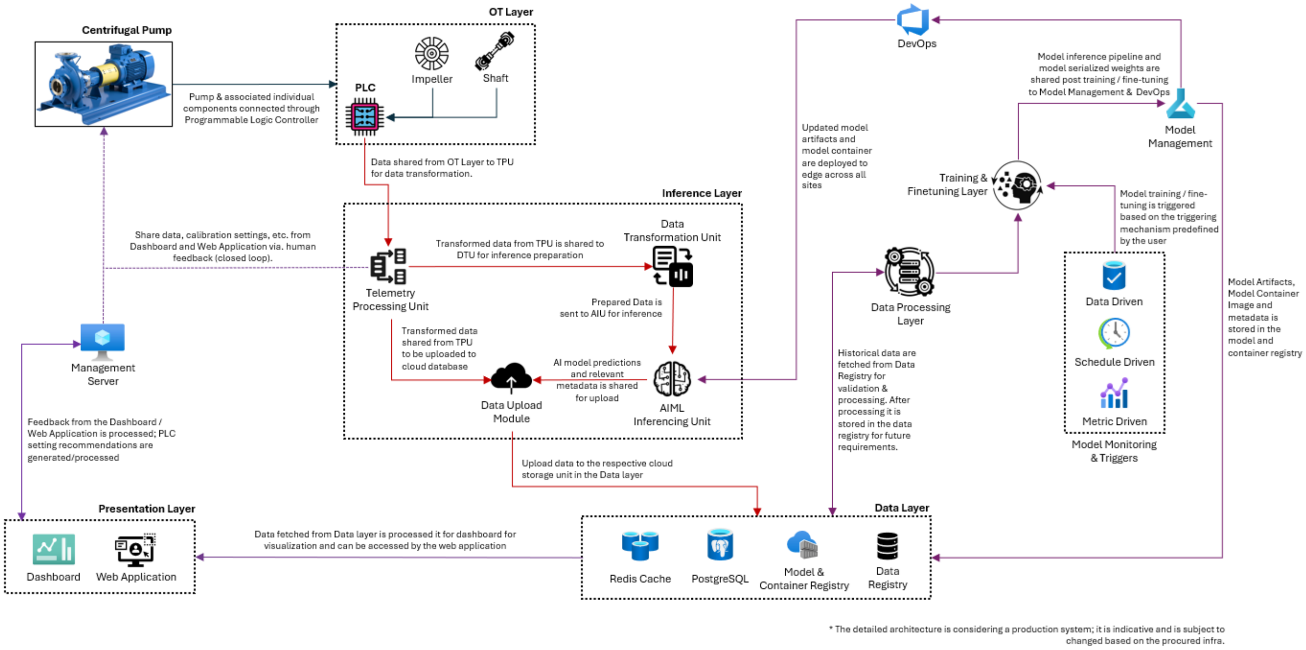


Figure 11: System Architecture Diagram

The proposed production system architecture diagram illustrates the flow of data and model artifacts across multiple integrated components, optimized for real-time operation with minimal latency. The process begins at the Centrifugal Pump, where data from associated components is collected via a Programmable Logic Controller (PLC) in the OT Layer. This data is transmitted to the Telemetry Processing Unit (TPU) within the Inference Layer for transformation. The Data Transformation Unit prepares the data for AI/ML inferencing, and the AIML Inferencing Unit produces model predictions, which are then shared with relevant metadata and uploaded to the cloud database using the Data Upload Module.

The Data Layer comprises sub-components such as Redis Cache, PostgreSQL, Model & Container Registry, and Data Registry, which store, manage, and provide access to processed data, model artifacts, and historical records. The Presentation Layer, containing Dashboard and Web Application sub-components, processes data for visualization and enables user feedback that is routed back to the Management Server for ongoing PLC calibration and closed-loop adjustment.

Model training, fine-tuning, and deployment are managed through the Training & Finetuning Layer, DevOps, and Model Management components. Model inference pipelines and serialized weights are deployed to server locations via DevOps, and model artifacts are stored in the registry. Triggers for model monitoring are data-driven, schedule-driven, or metric-driven, ensuring continuous performance and adaptation. Throughout the diagram, directional arrows convey the pathway and purpose of data exchanges, connecting each sub-component and major component as part of the real-time production pipeline.

20. Conclusion

This dissertation rigorously addresses the longstanding challenge of data scarcity and its ramifications for human-machine collaboration in complex industrial environments, through the deployment and evaluation of advanced generative AI techniques. By integrating robust anomaly detection with innovative synthetic data generation—leveraging Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and their conditional variants—the study delivers a comprehensive framework for enhancing both AI system reliability and human operational safety.

Extensive comparative analysis across six model evaluation tables highlights the inherent limitations of statistical and shallow learning techniques under severe class imbalance, with non-parametric approaches such as the Percentile method demonstrating moderate recall and F1-score but ultimately falling short in minority class discrimination. Traditional machine learning methods—most notably ensemble models such as Random Forest and XGBoost—achieve marked improvements in detection performance, particularly post-synthetic balancing, with SMOTE-based augmentation catalyzing optimal precision, recall, and F1-score. Notably, the Random Forest classifier consistently exhibits superior generalization and sensitivity in identifying rare anomalies, setting empirical benchmarks for model selection in industrial anomaly detection tasks. The deployment of deep learning methodologies—specifically autoencoders and VAEs—shows limited incremental value within the context of severe class imbalance, as evidenced by persistently low recall and F1-score metrics. Conversely, the application of class-conditional generative frameworks (CW-GAN) further advances data-driven anomaly synthesis, matching the efficacy of classical oversampling strategies and proving indispensable in scenarios marked by non-linear operational states and data heterogeneity.

Synthesizing these findings, it is evident that the integration of generative AI models—supported by rigorous feature engineering and strategically designed synthetic augmentation—empowers industrial workflows with unprecedented robustness and adaptability. The framework not only overcomes fundamental data limitations but also enables proactive anomaly detection, comprehensive operator training, and dynamic validation of AI-powered control systems. This multidimensional approach thus facilitates greater situational awareness, operational safety, and productivity, driving transformative advances in human-machine collaboration within Industry 4.0 environments. It establishes a compelling precedent and a robust platform for ongoing innovation, ensuring that both humans and machines can coalesce effectively amidst evolving operational complexities.

Looking ahead, innovations in GAN variants—such as improved architectures and multimodal industrial data—hold considerable promise for producing even higher-fidelity synthetic samples. As these generative models evolve, they are expected to bridge the gap between real and simulated operational scenarios, enabling traditional ML models to generalize more effectively amidst data complexity and imbalance. The continued development of these advanced generative frameworks will further amplify the reliability and applicability of AI-driven solutions in safety-critical industrial domains, marking a pivotal trajectory for research in collaborative intelligence.

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23. Check List

Sl No	Check list	Yes / No
1	Is the Cover page in proper format?	Yes
2	Is the Title page in proper format?	Yes
3	Is the Certificate from the Supervisor in proper format? Has it been signed?	Yes
4	Is Abstract included in the Report? Is it properly written?	Yes
5	Does the Table of Contents' page include chapter page numbers?	Yes
6	Is Introduction included in the report? Is it properly written?	Yes
7	Are the Pages numbered properly?	Yes
8	Are the Figures numbered properly?	Yes
9	Are the Tables numbered properly?	Yes
10	Are the Captions for the Figures and Tables proper?	Yes
11	Are the Appendices numbered?	Yes
12	Does the Report have Conclusions/ Recommendations of the work?	Yes
13	Are References/ Bibliography given in the Report?	Yes
14	Have the References been cited in the Report?	Yes
15	Is the citation of References/ Bibliography in proper format?	Yes

(Signature of Student)
Date: 17 August, 2025

(Signature of Supervisor)
Date: 17 August, 2025